

Clinical review

Malignant choroidal melanoma

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Abstract

Malignant choroidal melanoma is the most common primary intraocular malignancy in adults. Controversy has surrounded the treatment and management of these tumors for many years. The Collaborative Ocular Melanoma Study (COMS) is a randomized, controlled, clinical trial funded by the NEI. The study was established to answer the question: what type of treatment is most likely to cause tumor regression, and provide the patient with the best prognosis for cancer free survival? Patients with small tumors were followed annually for 2 years and again 5 years later to determine growth rates for these tumors. Patients with medium, and large tumors are randomized to various treatment strategies. These patients are followed bi-annually with a complete ophthalmic examination and once a year with a complete medical/oncological evaluation. The small tumor arm has completed enrollment with 188 patients. Thirty-one percent of these patients had significant growth of their tumor and were subsequently treated. Final conclusions from the medium tumor arm and the large tumor arm of the trial are pending. A case presentation of a patient with a medium size tumor and an update on the study design and the COMS data gathered to date will be discussed. © 1997 Elsevier Science Ireland Ltd.

Keywords: Choroidal melanoma; Enucleation; Brachytherapy; Metastasis; Study design

1. Case presentation

A 69-year-old white male was referred by his optometrist for evaluation of 'a shadow' seen in his left eye. The patient had no ocular complaints, his ocular and medical histories were unremarkable, and he was taking no medications. His family ocular history was remarkable for a maternal grandmother having 'glaucoma' and his mother having a retinal detachment in one eye.

Visual acuity was correctable to 20/15 OD and 20/20 OS with a slight hyperopic refraction, and J1 + at near in each eye with a +2.50 add. Confrontational fields were full to finger counting and extraocular motility was normal. Pupils were 5 mm, round, and briskly reactive to light with no afferent pupillary defect. The patient's anterior segments were clear and unremarkable except for very mild cataract for-

mation, OU. Applanation tonometry measured 12 mmHg in each eye. Dilated exam in the right eye showed a healthy optic nerve, vessels, macula and periphery. Dilated exam of the left eye revealed a healthy optic nerve, vessels, and macula. Peripherally at 4:00, there was an elevated, dome shaped, brown-green pigmented mass lesion with an area of lighter pigmentation superior-temporal and an overlying, localized, serous detachment of the retina (Fig. 1). Echographic evaluation revealed a basal diameter of 12 mm × 12 mm and an apical height of 4.2 mm (Fig. 2). The lesion showed low internal reflectivity and mild vascularity. No extraocular extension was noted (Fig. 3). A diagnosis of malignant melanoma of the choroid was made. The tumor was of medium size based on its clinical appearance and the echographic measurements.

Various treatment options were discussed with the patient including plaque radiotherapy, enucleation, and the opportunity to be enrolled in the Collaborative Ocular Melanoma Study (COMS). After careful

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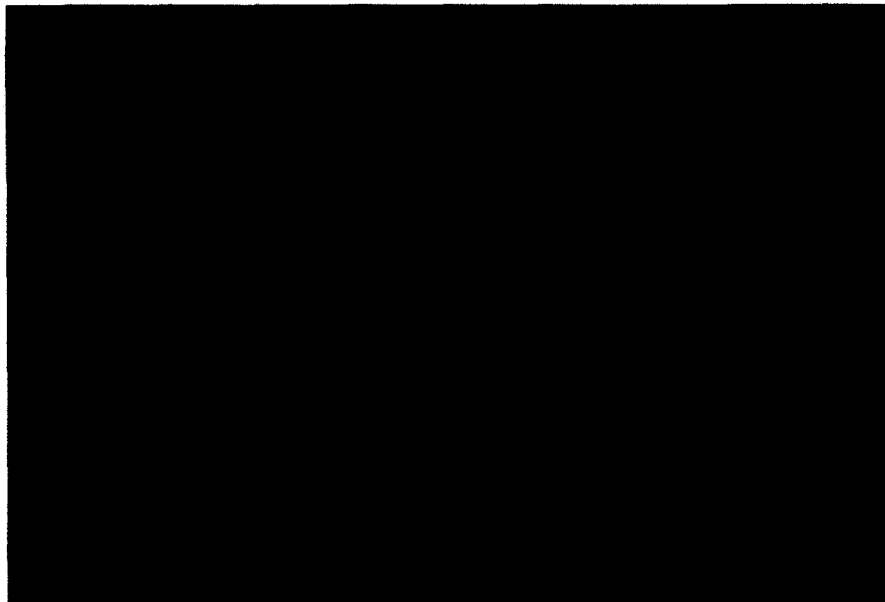


Fig. 1. Primary malignant choroidal melanoma in a 69-year-old white male.

consideration, the patient chose not to be enrolled in the study. He chose to receive plaque radiotherapy. The patient was referred to an Ocular Oncologist who confirmed the diagnosis. The patient underwent a complete medical and oncological evaluation including blood work, liver function tests, and chest X-ray. All tests were normal and the patient underwent surgery to apply a radioactive plaque containing radioactive Iodine (^{125}I). The plaque was left in place

for 3 days on an in-patient basis. After removal of the plaque, the patient was discharged on topical scopolamine ophthalmic solution one drop q.i.d. OS, prednisolone acetate ophthalmic solution one drop q.i.d. OS, and gentamicin ophthalmic solution one drop q.i.d. OS.

Upon follow-up 1 week later, the patient had a complaint of vertical diplopia only when looking down and to the left. His vision was correctable to 20/15



Fig. 2. B-scan of melanoma measuring 12 mm $\times$ 12 mm $\times$ 4.2 mm showing elevation, low vascularity, and adjacent retinal detachment.



Fig. 3. A-scan of melanoma showing height, and low internal reflectivity with no extraocular extension noted.

OD and 20/20 OS. In the left eye, there was 2 + injection of the bulbar conjunctiva. The cornea was clear and trace cells were noted in the anterior chamber. Applanation tonometry measured 10 mmHg in each eye. The tumor appeared unchanged and the rest of the left eye was stable. The diplopia was felt to be secondary to manipulation of the inferior rectus muscle when applying the radioactive plaque. The problem was resolved by adding an eight prism diopter, base-up, Fresnel prism to the patient's right bifocal segment. The gentamicin and scopolamine drops were discontinued and the prednisolone acetate ophthalmic solution was tapered.

The patient returned for follow-up 6 months later with no complaints. A hyper deviation was still present but was correctable with the Fresnel prism. All other examination findings were similar to the previous exam. Echography revealed no change in the size of the tumor and resolution of the retinal detachment. The patient was followed every 6 months thereafter. Although the patient was not part of the COMS, the same follow-up protocols were used. Examination 2 years after surgery showed the tumor had decreased in size. This was the first noted change in size since the treatment. Echography revealed a basal diameter of 10.5 mm by 11.5 mm and an apical height of 3.4 mm with no extra-ocular extension (Fig. 4). The internal reflectivity of the tumor was slightly increased which is characteristic of a regressing lesion status post-treatment (Fig. 5). Oncological evaluation revealed normal liver functions and normal chest X-ray.

The patient has had an excellent response to external brachytherapy and continues to remain stable.

1.1. Background

Choroidal melanoma represents the most common primary intraocular tumor in adults [1–3]. Peak incidence is in the early 60s representing about 7.5 cases per one million population. Prevalence is equal among males and females. Incidence is rare in younger adults under 30 years old with an estimated peak incidence of about six cases per one hundred million population. Caucasians are eight times more likely to develop the tumor than African-Americans and three times more likely than Asians [22,23].

Choroidal melanomas classically present as a brown, elevated, dome shaped, sub-retinal tumor with varying growth potential (Fig. 6). Pigmentation can range from dark brown to yellow-white in color. The lighter 'amelanotic' tumors represent approximately one-third of all melanomas (Fig. 7). Visual symptoms may vary with peripheral lesions. Patients may be asymptomatic and unaware of a lesion or may complain of flashes of light secondary to the overlying serous retinal detachment. However, a more centrally located lesion in the posterior pole may cause a decrease in vision and/or metamorphopsia. Geographic areas of orange pigment can be seen overlying some lesions. Histopathologically, these areas consist of macrophages containing lipofuscin and melanin granules from degenerated retinal pigment epithelial cells [4,28]. Progressively

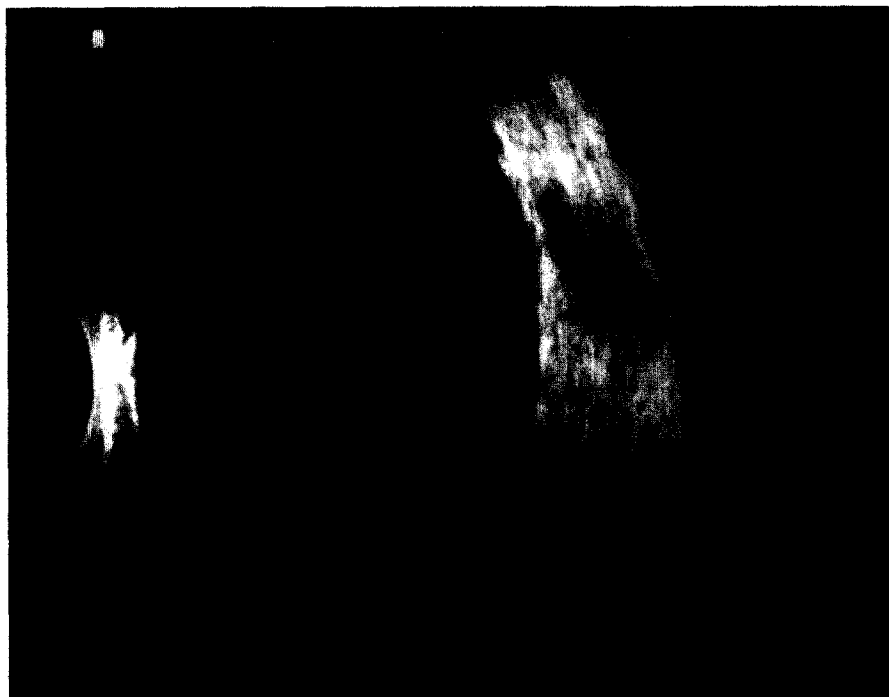


Fig. 4. B-scan of melanoma status post ^{125}I brachytherapy showing decreased elevation, no retinal detachment, and increased internal reflectivity.

growing tumors acquire a classic 'mushroom' shape after they have broken through Bruch's membrane. Choroidal neovascular membranes are occasionally seen overlying a tumor and this can lead to hemorrhage or an exudative retinal detachment. A serous retinal detachment may also occur over a melanoma.

Because choroidal melanomas can have such a large variation in their appearance, ancillary testing plays a very important role in diagnosing, treating and following these tumors. According to Murray, 'The most important tool of diagnosis is the binocular indirect ophthalmoscopic examination. This allows the clini-

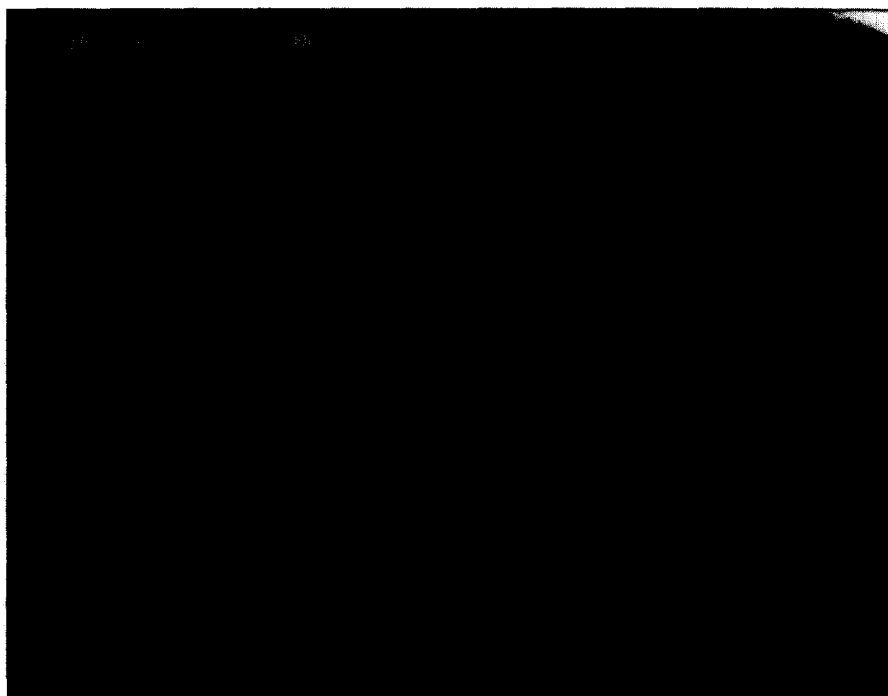


Fig. 5. A-scan of melanoma status post ^{125}I brachytherapy showing decreased elevation and increased reflectivity.

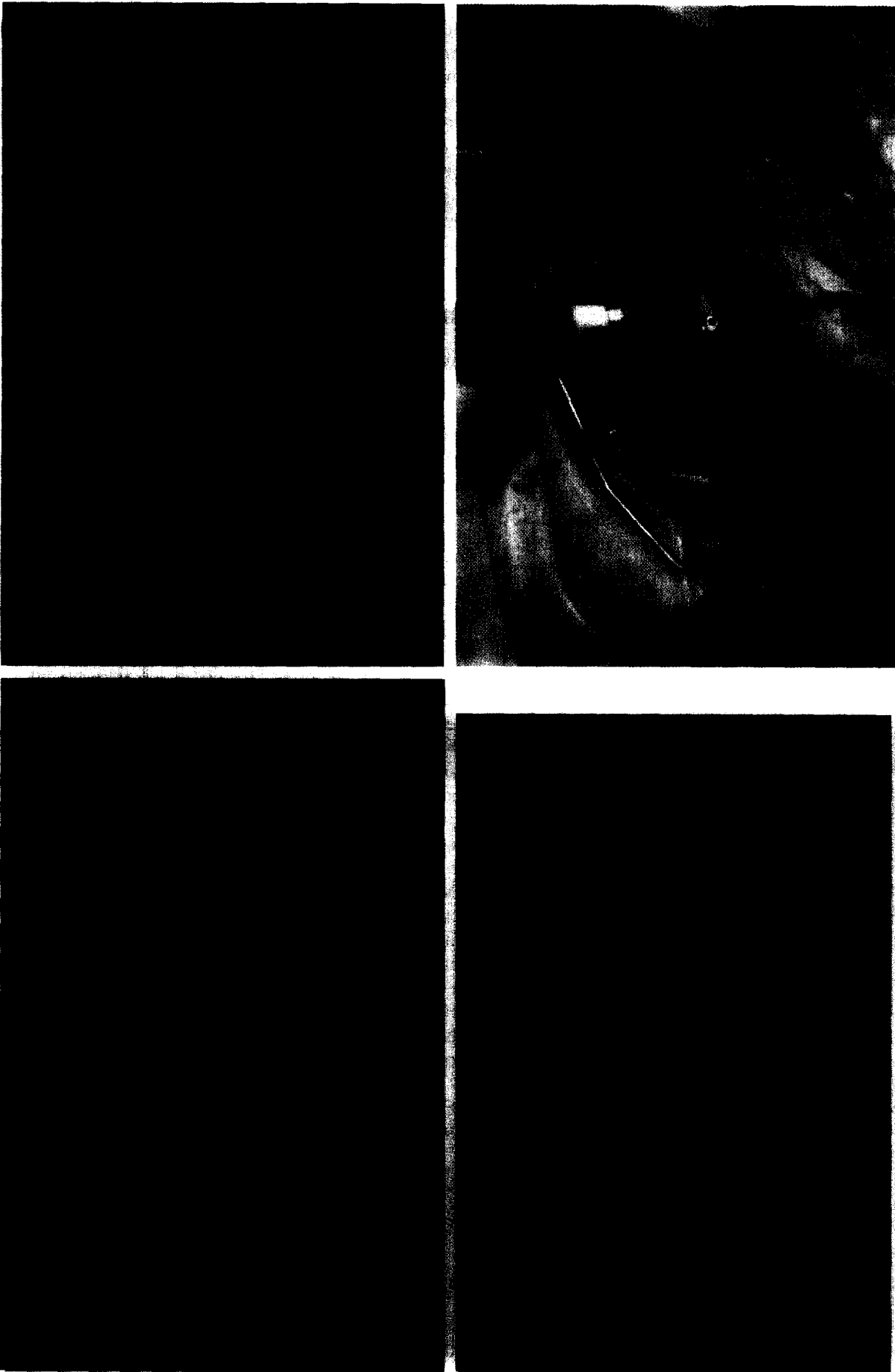


Fig. 6. (*top left*) Large malignant melanoma showing characteristic dome shape and brown/green appearance in a 64-year-old white male.

Fig. 7. (*top right*) Typical amelanotic melanoma of the posterior pole in a 72-year-old white female.

Fig. 8. (*bottom left*) Transillumination of choroidal melanoma showing extension of the tumor borders.

Fig. 9. (*bottom right*) Intraoperative outline of tumor base on sclera with a marking pen.

cian to determine the size, shape, color, and precise anatomical location of the lesion' [3]. Another simple yet important test that can be performed in the office is transillumination of the tumor. Transillumination can give the clinician important information on the tumor size, density, and extent of its growth. There are three different ways to transilluminate a tumor. The transcleral technique places the light source directly on the sclera over the tumor location and the opposite side of the eye is observed. In the transpupillary technique, the light source is placed directly in the conjunctival fornix directed toward the sclera. The amount of light transmitted through the pupil is observed. In the indirect ophthalmoscopic technique, the lesion is localized with the BIO. A bright light source is then placed over the base of the tumor on the sclera and the light of the BIO is turned off. This allows localization of the tumor by viewing the transillumination. Based on how the lesion transilluminates, the examiner can rule-out other lesions such as retinal detachment and serous detachments of the choroid. In all of these techniques, a pigmented tumor or blood will block the illumination while a retinal detachment or a non-pigmented lesion will allow the illumination through (Fig. 8) [3,8]. Baseline fundus photographs are important on the first visit to document change in tumor size over time. Fluorescein angiography is not diagnostic for choroidal melanoma, but plays a role in helping to rule-out other lesions such as choroidal neovascular membranes, macroaneurysms, or vascular tumors. Typical fluorescein patterns change with the size and extent of the tumor. A small tumor may not reveal any abnormal staining patterns while a larger tumor will show some pinpoint hyperfluorescence in the venous phase. Orange pigment may either stay hypofluorescent throughout or become slightly hyperfluorescent in the later stages. Large dome-shaped tumors show highly characteristic features. In the arterial and early venous phases the prominent choroidal vessels in the tumor show relative hyperfluorescence. Simultaneously the retinal vessels hyperfluoresce giving a characteristic 'double circulation' pattern to the angiogram.

The most important test used to diagnose and follow these lesions is standardized echography. This test effectively evaluates tumors greater than 1 mm in apical height. In the hands of a skilled echographer, changes in tumor height of greater than or equal to 0.3 mm can be recognized. The standardized A-scan allows for tissue differentiation. The B-scan can also give important information on vascularity, reflectivity, and extra-ocular extension of the tumor which can be helpful in establishing a differential diagnosis for suspicious lesions. Typically choroidal melanomas tend to be of low to medium internal reflectivity and have

a low to medium amount of vascularity within the tumor [30].

A controversial test used to differentiate malignant from benign lesions is the radioactive Phosphorous (^{32}P) uptake test. In this test, radioactive ^{32}P is injected transclerally directly into the lesion. Radioactivity is then measured in the lesion. A malignant lesion will readily show uptake of the ^{32}P where a benign lesion will not. This test is usually reserved for those lesions in which a concrete diagnosis cannot be made by non-invasive techniques.

2. Differential diagnosis

When presented with a suspicious uveal lesion, the clinician must first rule-out other causes before making a diagnosis of malignant melanoma. Choroidal nevi are the most common lesions misdiagnosed as melanomas. The typical choroidal nevus is flat, slate gray in color, and 95% are less than two disc diameters in size and easily distinguishable from a choroidal melanoma. Suspicious nevi tend to be larger than three disc diameters and slightly elevated. Features that may be seen in a suspicious choroidal nevus and can help differentiate it from a melanoma include high internal reflectivity on echographic evaluation, drusen, choroidal neovascular membranes, pigment migration, and pigment atrophy. If a lesion has lipofuscin overlying it, this may signify that it is more likely to have malignant potential and should be followed closely with serial photographs and echography. Standard strategies to manage patients with suspicious nevi are summarized in Table 1. It is felt by most clinicians that there is no harm to the patient in observing suspicious nevi or suspected small tumors for growth [5].

Malignant melanomas can also be confused with a localized hemorrhage under the retinal pigmented epithelium, congenital hypertrophy of the retinal pigment epithelium (CHRPE), choroidal hemangioma, uveal effusion syndrome, serous choroidal detachments, and rhegmatogenous retinal detachments. Localized hemorrhages under the RPE secondary to a choroidal neovascular membrane tend to block fluo-

Table 1
Suggested follow-up criteria for suspicious choroidal nevi

Height of lesion	Suggested follow-up	Ancillary testing
< 2 mm	1 year	Baseline photographs
2–3 mm	Every 3 months	Photographs, echography, baseline fluorescein angiogram
> 3 mm	Every 3 months or treatment if indicated.	Photographs, echography, baseline fluorescein angiogram or treatment if indicated.

rescence early in the angiogram and the membrane diffusely stains later. Fresh hemorrhage will have a low internal reflectivity on echography and disciform lesions will have high internal reflectivity.

Congenital Hypertrophy of the Retinal Pigment Epithelium lesions are sometimes mistaken for melanomas, but these lesions are always flat and have well circumscribed margins.

Choroidal hemangioma, choroidal osteoma, and metastatic lesions can be mistaken for an amelanotic melanoma. Choroidal hemangiomas typically have a deeper red color in their clinical appearance and characteristic fluorescein staining patterns with early hyperfluorescence and high internal reflectivity on echography [29]. Choroidal osteomas have a higher incidence in females, are localized around the optic nerve and have a high internal reflectivity. Metastatic lesions tend to be amelanotic, multiple, bilateral, relatively flat, rarely mushroom-shaped, and with more sub-retinal fluid than that seen in melanomas.

Uveal effusion syndrome is a rare, idiopathic condition characterized by annular choroidal detachment associated with shifting subretinal fluid. However, it shows no solid component to the mass, transmits light with transillumination, and the associated retinal detachment is more extensive than with a melanoma of similar size [29].

Serous choroidal detachments typically are multilobular in appearance and usually follow ocular surgery while the eye is in a state of hypotony. The color and elevation may look similar to a pigmented melanoma. However, these lesions will transmit light on transillumination. A more difficult diagnostic problem may develop with a hemorrhagic choroidal detachment. In these cases the blood will block light on transillumination and may show low to medium internal reflectivity if the blood is fresh. If this is the case, further testing is necessary to rule out a melanoma such as the ^{32}P uptake test. In a melanoma this test will be positive.

Rhegmatogenous retinal detachments have been mistaken for melanomas in numerous studies leading to enucleation. In contrast to a melanoma, the fluid associated with a rhegmatogenous detachment does not shift with movement of the patient's head, the overlying retina has a rippled surface appearance, and a retinal hole can usually be localized. Transillumination would allow transmission of light in a detachment of this kind.

3. Treatment

Various treatment options exist depending on the size, extent, and location of the tumor. Enucleation had been the standard of treatment for choroidal melanomas since the 1800s. With the recent advances

in technology, other treatment options are now available which allow the patient to preserve their eye, their sight, and prolong their life. Other factors taken into consideration when deciding upon a treatment strategy include status of the fellow eye, the age of the patient, the general health of the patient, and the psychological status of the patient.

Small melanomas, which are less than 2.5 mm in apical height and less than 10 mm in basal diameter, have the largest variation in treatment options. Gass wrote that it is safe to observe small tumors for growth prior to enucleation [5]. Many clinicians elect to observe these lesions every 3–4 months until there is evidence of rapid growth before deciding on a more aggressive approach. Laser photocoagulation is used by some clinicians in rare cases to stop the mitotic activity of the tumor. This option is usually reserved for small tumors in a location not favorable to radiation, i.e. the peripapillary area. The biggest disadvantage to this option is the need for multiple treatment sessions to effectively treat the lesion. Also, even when the lesion may seem to be adequately treated, there may be inadequate penetration by the laser to the base of the tumor. Surgical excision is usually reserved for small tumors of the ciliary body and equatorial choroid [9]. The main argument against surgery is the technical difficulty of the procedure and the high rate of complications.

Medium sized tumors measure 2.5–10 mm in apical height and less than 16 mm in basal diameter. There are currently two treatment options accepted for these tumors; plaque radiotherapy and enucleation. Enucleation was challenged by Lorenz E. Zimmerman in 1978 when he claimed that manipulation of the eye with a uveal melanoma actively disseminated tumor emboli into the blood stream. In approximately 66% of the cases he followed, there was an increase in the mortality rate from 1 to 8 % peaking 2 years after the enucleation [14,15]. This thought process has been disputed in recent studies. It is felt that tumor growth led to enucleation of the eyes in the study, thus, it would be reasonable to blame the metastasis on the tumor growth rather than on the treatment [20].

Plaque radiotherapy consists of a surgical procedure where a plaque of radioactive material is sutured to the sclera overlying the tumor and left in place for 3 days. ^{60}Co was the first isotope used because of its high energy disposition to the tumor and its availability [26]. Recently, however, ^{125}I has become the preferred choice because the radiation dose to the tumor is similar to ^{60}Co but the dose to the globe opposite the plaque is less than other radioactive materials. The plaque is contained in a solid gold housing to protect the other ocular structures. The solid gold shielding used with ^{125}I is safer for other structures in the eye as well as all personnel involved in plaque

preparation and placement. Such shielding cannot be accomplished with the higher energy radioisotopes such as ^{60}Co [10–13,26]. Enucleation of medium size tumors is generally indicated for tumors that are too close to or have invaded the optic nerve. It is believed that radiotherapy results in profound vision loss in these cases and the degree of invasion of the nerve is difficult to assess with ophthalmoscopy, echography, or computed tomography [9].

Large tumors are considered to be greater than 10 mm in height and greater than 16 mm in basal diameter. These tumors tend to be visually threatening because of the invasive nature of a large tumor on the macula and optic nerve. Enucleation and Pre-enucleation External Beam Radiotherapy (PERT) followed by enucleation are the two most common treatment options. It is believed that external beam radiotherapy decreases the mitotic activity of the tumor prior to enucleation thus, decreasing the chance of metastasis [16–18]. Char and associates disputed this idea in their study of 41 patients [19]. They concluded that pre-enucleation radiotherapy failed to prolong patient survival because of the existence of micro metastasis before treatment. However, their study was retrospective and had a small sample size. Their conclusions should be regarded as tentative [18,20]

4. Discussion

In 1986, The Collaborative Ocular Melanoma Study was started to address the controversy regarding the management of choroidal melanomas. The study represents the largest NEI trial and has continued funding through the year 2000. Currently 43 centers are enrolling patients and as of the summer of 1995, over 6000 patients had been enlisted. This represents over

half of all diagnosed cases of posterior uveal melanoma within North America. The COMS is the only randomized, multicenter, prospective trial evaluating treatment of patients with posterior uveal melanoma [1,7].

The criteria for inclusion and exclusion in the study are outlined in (Table 2) [1,7]. Patients enrolled in the trial are randomized into one of the treatment modalities depending on the tumor classification. Medium sized tumors are randomized to enucleation or external brachytherapy. Large tumors are randomized to enucleation or external beam radiotherapy followed by enucleation. The large tumor trial has completed enrollment with 1003 patients randomized. A total of 204 patients were enrolled in the small tumor trial [2]. And the medium tumor trial continues to enroll patients with an enrollment goal of 2400 patients [3].

The primary outcome for all arms of the trial is 'all cause' mortality. All-cause mortality rates are used to minimize report selection and data extraction biases. It is believed that earlier reports estimating mortality rates for melanoma are flawed because these reports did not differentiate causes of mortality in patients. It may be assumed that all patients died from metastasis, when in fact, they may have died from other causes not related to their melanoma. It is estimated that 5-year all-cause mortality depends on the size of the tumor at the time of treatment and is a major prognostic factor for patients with choroidal melanoma. A study by Diener–West and associates found that tumor size at the time of treatment was the major factor for patient mortality [21]. Small tumors had a 16% 5-year mortality rate versus 32% for medium sized tumors and 56% for large tumors following enucleation [21]. Secondary outcomes for all arms of the study are listed in (Table 3) [1,7]. Several

Table 2
COMS eligibility and exclusion criteria

General eligibility requirements	Exclusion criteria
1. Primary tumor in only one eye.	1. Any social, mental, or physical condition that impairs ability either to give informed consent or to return to the clinic for regular follow-up visits.
2. Accurate delineation of tumor size possible with echograms.	2. Use of immunosuppressive therapy that might alter the body's response to the melanoma.
3. 21 years of age or older.	3. Pregnancy or breast-feeding.
4. Able to give informed consent.	4. Previous intervention for choroidal or ciliary body melanoma.
5. Able to return for follow-up visits.	5. Previous intervention in the eye for any condition secondary to the tumor.
	6. Previous fine-needle aspiration biopsy.
	7. Gross extra scleral extension of the melanoma.
	8. Diffuse 'ring' melanoma or more than one melanoma in the given eye.
	9. Other known primary or metastatic malignancy except non-melanotic skin cancers and carcinoma in situ of the uterine cervix.
	10. One half or more of the tumor involving the ciliary body.

Table 3
Synopsis of outcomes for the COMS

Primary outcome	Secondary outcomes
Small tumors	
All-cause mortality	Growth rates Metastasis
Medium tumors	
All-cause mortality	Metastasis Cancer related mortality Vision retention in treated eye Radiation complications
Large tumors	
All-cause mortality	Metastasis Cancer related mortality Vision retention in fellow eye

ancillary studies are ongoing within the COMS including the evaluation of histopathological features of all enucleated eyes, the accuracy of diagnosis in patients undergoing enucleation, the characterization of demographics, and clinical signs and symptoms for all patients.

Diagnostic accuracy in the trial has been excellent at approximately 99.5% [6]. This number has been steadily increasing over time presumably because of better diagnostic techniques and more clinical experience. Previous studies have placed diagnostic accuracy from 92% to 98% [6]. Tumor growth assessment is done bi-annually. Complete oncological work-up is done yearly. All-cause mortality rates are used to evaluate death rates among patients in the study regardless of management criteria the patient was under.

A small tumor trial evaluating tumors with an apical height between 1 and 3 mm and a basal diameter of ≥ 5 mm was instituted from 1986 to 1989. These tumors encompassed 30% of tumors evaluated in the COMS within this period. The patients were followed with baseline echography, fundus photographs, and fluorescein angiogram studies along with complete

demographic data for a period of 2 years annually. A further follow-up round of data was collected in 1994 to determine tumor growth. Fifty-eight of the 188 patients (31%) had significant tumor growth [3]. All of these patients were treated for growth. There was an average growth rate of 10% per year. The 5-year Kaplan–Meyer estimate of all-cause mortality for small tumors was 6% [3].

The medium sized tumor trial encompasses tumors with an apical height of 2.5 mm to ≤ 10 mm and a basal diameter of ≤ 16 mm. For peripheral lesions the tumor cannot involve more than 50% of the ciliary body. Posterior lesions must be ≥ 2 mm from the optic nerve. All patients undergo complete ophthalmic and metastatic evaluation. Patients are also seen in consultation with a radiation oncologist prior to treatment randomization. These patients are randomized to standard enucleation therapy or external plaque radiotherapy with ^{125}I . Prior to radioactive plaque placement the tumor base is outlined on the sclera using a marker (Figs. 9 and 10). The radioactive plaque is then placed in a solid gold housing resembling a bottle cap (Fig. 11). The plaque is then sutured to the sclera and left in place for 3 days (Fig. 12). After removal of the plaque, the patient is followed immediately following surgery and twice yearly with a complete ophthalmic evaluation. A complete medical/oncological evaluation is done yearly.

In the large tumor trial, tumors were classified as being large if they were ≥ 10 mm in apical height and ≥ 16 mm in basal diameter. Tumors with an apical height of ≥ 8 mm and in a peripapillary location were included in this trial. Less than 50% of ciliary body involvement was required for enrollment. Patients were randomized to standard enucleation therapy or PERT followed by standard enucleation [1,3,7]. The rationale behind this arm of the study is to determine if PERT will help to decrease the mitotic activity of the tumor prior to enucleation thus decreasing the chance of metastasis [17,24,25]. Follow-up is scheduled for no less than 10 years or until death (Table 4).

The most common site of metastasis from a primary choroidal melanoma is the liver. The second most common site is the lungs. Overall, it is estimated that about 30–40% of patients develop evidence of metastatic disease within 10 years after removal of the eye [21]. One study of 5-year survival rates following enucleation found mortality rates of 16% for small tumors, 32% for medium tumors, and 53% for large tumors [21,27].

5. Conclusion

Melanoma represents one of the few ocular conditions that can be potentially life threatening and

Table 4
COMS treatment arms

Definition	Treatment arms
Small tumors	
Apical height between 1 and 3 mm and a basal diameter ≥ 5 mm.	Observation annually for 2 years. Further follow-up data obtained 5 years later documenting growth.
Medium tumors	
Apical height of 2.5 mm to ≤ 10 mm and a basal diameter of ≤ 16 mm.	Randomized to standard enucleation therapy or external plaque radiotherapy.
Large tumors	
Apical height of ≥ 10 mm and a basal diameter of ≥ 16 mm.	Randomized to standard enucleation therapy or PERT followed by enucleation.

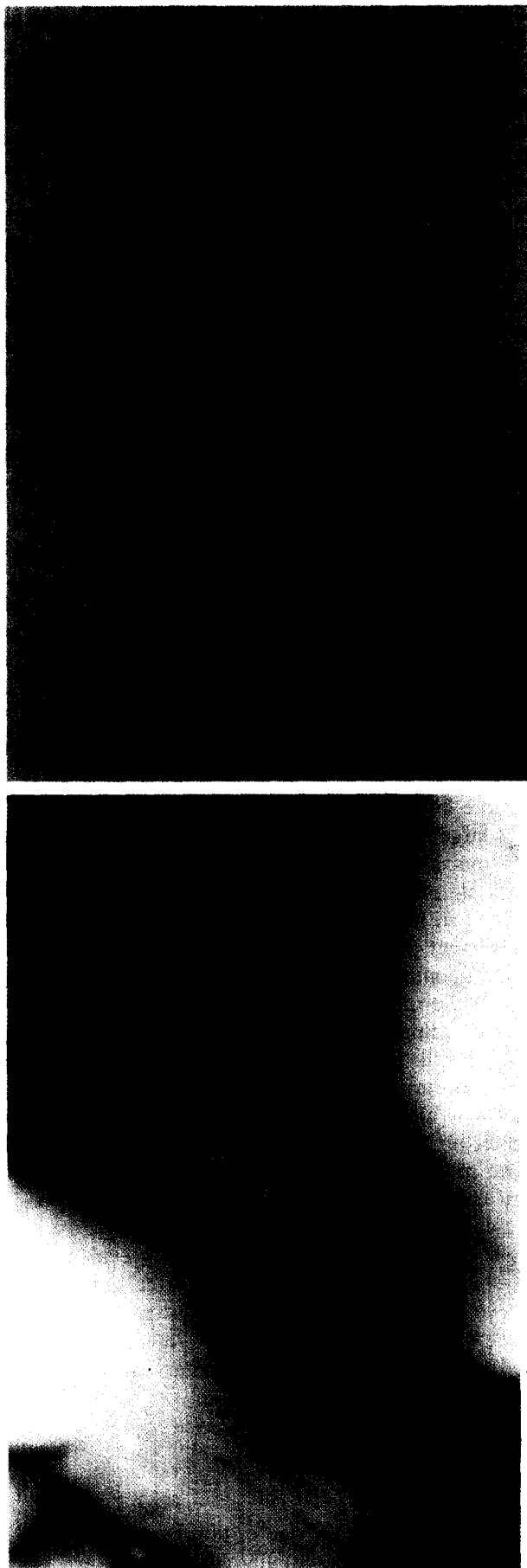


Fig. 10. (top left) Measurement of tumor base to verify plaque size and placement intraoperatively.

Fig. 11. (top right) Solid gold plaque housing with 13 radioactive ^{125}I seeds inside.

Fig. 12. (bottom) Intraoperative photo of gold housing and ^{125}I plaque sutured to sclera.

requires appropriate and efficient management. Controversy still exists regarding what the proper timing and method of treatment is for this rare condition. A large, prospective, randomized, controlled clinical trial is currently in progress to answer these management questions. The Collaborative Ocular Melanoma Study is nearing completion of its enrollment and will help to give clinicians direction when deciding the best treatment option for their patients. For now, the most important aspect of management is to be able to recognize the characteristics of a malignant melanoma, make the appropriate diagnosis, and make an appropriate referral to a qualified ocular oncology specialist.

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